Ex:no: 01
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NETWORK COMMANDS

AIM:

To study various network commands used in Linux and windows

PROCEDURE:

1. open the command prompt / Terminal.
2. Execute the required network commands
3. observe and record the output of each command.
4. Analyse the network information displayed by the commands.

COMMANDS:

1. ARP Command
   ~ arp -a
   Purpose: Display IP and MAC address information

2. Hostname:
   ~ hostname
   Purpose: Display name of the Computer

3. IP Configuration
   ~ ipconfig /all
   Purpose: Display detailed TCP/IP Configuration information

4. NetBIOS
   ~ netstat -a
   Purpose: Displays network connections and routing info

5. Network Statistics:
   ~ netstat -r
   Purpose: Displays NetBIOS name information

6. DNS Lookup:
   ~ nslookup www.google.com.
   Purpose: Display DNS information of the Specified domain

C:\> arp -a

Interface: 192.168.1.10

Internet Address	Physical Address	Type
192.168.1.1	00-11-22-33-44-55	dy

C:\>hostname
DESKTOP-PC

C:\> ipconfig /all
IPv4 Address: 192.168.1.10
Default Gateway: 192.168.1.1
DNS Server: 8.8.8.8

C:\> net stat -r

Network destination	Gateway
0.0.0.0	192.168.1.1

C:\> nslookup www.google.com.
Name: www.google.com
Address: 142.250.195.68

C:\> ping google.com
Reply from 142.250.195.68: bytes=32 time=20
TTL=
Reply from 142.250.195.68. bytes=32 time=16ms

7. Ping:

~ping google.com

Purpose: checks Connectivity between the local System and a destination host

8. Route

~route.

Purpose: Displays or modifies the IP routing table.

9. IP Command.

~ip address show

Purpose: Display IP address assigned to network interfaces.

10. MTR:

-mtr google.com

Purpose: Displays the route and network statistics for each hop.

11. TCPDump

~tcpdump -i eth0

Purpose: Captures and displays network packets.

12. NMCLI

~nmcli Connection show

Purpose: display Network Manager Connection profiles

Result:

Thus, the command to run network configuration, IP address, MAC address, routing information, DNS details, and Connectivity status were displayed Successfully.